

AK VisionFlow — Phase 6

Claude Development Prompt
Multi-Branch + Warehouse + Advanced Business Control

Critical context: Phase 1–5 are existing/planned foundation phases. Phase 6 is an incremental upgrade to the existing AK VisionFlow repository. Inspect the current implementation first. Do not rebuild working modules, reset the database, or delete/corrupt existing tenant data.

1. Mission

Expand AK VisionFlow from a single-location capable business system into a robust multi-branch and multi-warehouse platform. Give owners centralized control over branches, warehouses, stock movement, purchasing, sales, expenses, cash, performance and approvals while preserving the existing tenant-isolated architecture.

2. Non-Negotiable Rules

- Preserve all existing Phase 1–5 functionality.
- Never reset, truncate, recreate or destructively migrate the production database.
- All schema changes must use safe, versioned migrations.
- Existing tenants, branches, users, products, inventory, sales, purchases, accounting records, customers, suppliers, optical orders and payments must remain intact.
- Every branch and warehouse record must belong to the correct tenant.
- Every branch-scoped transaction must enforce branch authorization at the backend.
- Maintain backend-authoritative financial and inventory calculations.
- Use transaction-safe and idempotent stock-transfer operations.
- Preserve offline-first operation and cloud synchronization.
- Do not hard-code any single business or branch.
- Run the full existing test suite and add comprehensive Phase 6 regression tests.

3. Organization Structure

Tenant / Organization → Branches → Warehouses / Stores → Departments / Counters → Users

Support businesses with one branch or many branches. The same architecture must support a central/head office, retail branches, optical laboratories and central warehouses without forcing every customer to use every layer.

4. Branch Management

- Create, edit, activate/deactivate branches.
- Branch name, code, address, contact details and business configuration.
- Branch-specific operating settings.
- Assign users to branches.
- Define user access to one or multiple branches according to permissions.
- Branch-specific sales, purchases, expenses, payments and reports.
- Branch opening/closing status where operationally useful.
- Branch-wise cash management.
- Branch-wise performance comparison.

- Branch-level audit trail.
- Safe handling of inactive/closed branches without deleting historical data.

5. Warehouse Management

- Create and manage multiple warehouses/stores per tenant.
- Associate warehouses with branches or central/head-office operations.
- Warehouse stock visibility.
- Warehouse-wise inventory valuation.
- Stock receiving.
- Stock dispatch.
- Stock transfer.
- Stock adjustment with approval controls.
- Warehouse activity/history.
- Support central warehouse → branch replenishment.
- Prevent unauthorized cross-branch stock visibility.

6. Inter-Branch / Inter-Warehouse Stock Transfers

Transfer Request → Approval (where required) → Dispatch → In Transit → Receive → Completed

- Transfer request creation.
- Source and destination branch/warehouse.
- Product, quantity, batch/serial information where applicable.
- Transfer approval thresholds.
- Dispatch confirmation.
- In-transit status.
- Receiving confirmation.
- Short/damaged/rejected receiving.
- Automatic source stock reduction only at the correct dispatch state.
- Automatic destination stock increase only at the correct receiving state.
- Complete transfer history and audit trail.
- Idempotency so offline retries cannot duplicate stock.

7. Inventory Control Across Locations

- View total stock across all branches and warehouses.
- View available, reserved, in-transit and damaged quantities where supported.
- Branch/warehouse stock cards.
- Location-wise stock valuation.
- Stock movement ledger.
- Location-wise low-stock thresholds.
- Location-wise reorder levels.
- Central replenishment recommendations.

- Stock aging by location.
- Slow-moving/dead stock by location.
- Expiry monitoring by location for medicine inventory.
- Drill-down from inventory dashboard to the exact location and transaction.

8. Central Purchasing & Replenishment

- Allow central purchasing for multiple branches.
- Consolidate branch stock requirements.
- Generate replenishment recommendations.
- Create purchase requests based on location-level stock.
- Support central warehouse receiving.
- Distribute received stock to branches through controlled transfers.
- Track purchasing source and destination usage.
- Compare branch demand and consumption.

9. Branch-Wise Financial Control

- Branch-wise sales.
- Branch-wise gross profit.
- Branch-wise expenses.
- Branch-wise cash.
- Branch-wise bank activity where applicable.
- Branch-wise receivables.
- Branch-wise payables.
- Branch-wise purchase cost.
- Branch-wise inventory value.
- Branch-wise Profit & Loss.
- Consolidated organization-level financial reporting.
- Clear handling of inter-branch transfers so they do not create false external revenue.

10. Advanced Business Command Center Integration

Extend the Phase 4 Advanced Admin/Owner Business Command Center so the owner can switch between consolidated and location-level views without leaving the main dashboard.

- Global / All Branches view.
- Single Branch view.
- Branch comparison.
- Warehouse comparison.
- Sales by branch.
- Profit by branch.
- Inventory value by branch/warehouse.
- Low-stock locations.

- Excess/slow-moving stock locations.
- Inter-branch transfer pipeline.
- Pending transfer approvals.
- Branch expenses.
- Branch receivables/payables.
- Top and underperforming locations.
- Drill-down from every metric to source records.

11. Staff & Business Control — Lightweight, Not Full HR

AK VisionFlow is primarily targeted at shops and small/medium healthcare businesses, so do not build a heavy HR/payroll system in Phase 6. Implement only the operational staff-control features that directly improve business management.

- User-to-branch assignment.
- Role and permission control.
- Sales by staff.
- Discounts by staff.
- Returns/reversals by staff.
- Cashier shift/open-close control where appropriate.
- Basic attendance/presence capability only if it fits the existing architecture.
- Staff activity and audit history.
- Staff performance reporting.
- Commission/incentive tracking if compatible with the accounting design.

12. Approval & Control Workflows

- Purchase approval.
- Large expense approval.
- Large discount approval.
- Stock adjustment approval.
- Inter-branch transfer approval.
- Manual accounting adjustment approval.
- Configurable thresholds by tenant where practical.
- Approval history: requester, approver, date/time, decision and reason.
- Backend authorization must enforce approvals; UI-only restrictions are insufficient.

13. Branch-Level Permissions

- User can be assigned to one or multiple branches.
- Users should only see data within their authorized scope.
- Managers may have branch-wide permissions.
- Tenant Admin/Owner may access all branches.
- Cross-branch reporting must be permission-aware.

- Backend must enforce branch scope on every relevant API.
- Offline data must also respect the user's branch/tenant authorization.

14. Offline & Sync Requirements

- Support offline stock-transfer requests where the existing sync architecture permits.
- Use idempotency keys/unique operation identifiers.
- Prevent duplicate dispatch, receiving or stock updates after retry.
- Handle conflicts deterministically and safely.
- Never allow offline actions to bypass tenant/branch permissions.
- Display sync status and unresolved conflicts clearly.
- Preserve existing sync queue and retry behavior.

15. Reporting

- Branch Sales Report.
- Branch Profitability Report.
- Branch Expense Report.
- Branch Inventory Report.
- Warehouse Stock Report.
- Stock Transfer Report.
- Stock Movement by Location.
- Central Purchasing Report.
- Branch-wise Receivables/Payables.
- Branch comparison dashboard/report.
- Location-wise slow-moving/dead stock.
- Location-wise expiry risk.

16. UX Requirements

- Keep the system simple for small shops while supporting multi-branch enterprises.
- Do not force users to configure unnecessary warehouses or departments.
- Use clear location selectors and breadcrumbs.
- Always show current branch/warehouse context where relevant.
- Use clear transfer statuses.
- Provide search, filters and pagination for large datasets.
- Maintain responsive/mobile/PWA compatibility.
- Preserve existing light/dark mode and design language.
- Show useful empty/loading/error states.

17. Testing Requirements

- Test tenant isolation across branches and warehouses.
- Test branch-level RBAC at backend level.

- Test single-branch and multi-branch tenants.
- Test stock transfer lifecycle end-to-end.
- Test partial/short/damaged receiving.
- Test offline retry and duplicate prevention.
- Test inventory quantities and valuation before/after transfers.
- Test branch P&L; and consolidated reporting.
- Test inter-branch transfers do not create false sales/revenue.
- Test approvals and unauthorized actions.
- Test migration against realistic existing data.
- Run the complete regression suite from earlier phases.

18. Migration & Data Safety

Inspect the existing branch, inventory, sales, purchase and accounting schema before adding location hierarchy. If the existing system already contains a branch concept, extend it rather than duplicating it. If existing inventory is not location-scoped, create a safe migration strategy that assigns existing stock to an explicitly chosen/default location without silently losing quantities. Never fabricate historical branch performance.

19. Required Deliverables

- Multi-branch management.
- Multi-warehouse management.
- Location-aware inventory.
- Inter-branch/inter-warehouse stock transfer workflow.
- Central purchasing/replenishment support.
- Branch-wise financial/business reporting.
- Lightweight staff/business controls (not full HR).
- Approval workflows.
- Branch-aware Advanced Command Center.
- Safe database migrations.
- Automated regression tests.
- Updated documentation and deployment notes.
- Phase 6 completion report.

20. Acceptance Criteria

- Existing Phase 1–5 functionality remains intact.
- No existing data is deleted, reset or corrupted.
- A tenant can operate one or many branches.
- A tenant can operate one or many warehouses/stores.
- Stock quantities remain accurate across locations.
- Transfers are controlled, auditable and idempotent.
- Branch permissions are enforced server-side.
- Branch-wise and consolidated reports are accurate.

- Advanced Command Center supports global and branch-level views.
- Approval workflows cannot be bypassed through direct API calls.
- Offline/sync behavior remains safe.
- All existing tests plus new Phase 6 tests pass.
- No critical security or data-integrity issue remains open.

21. Claude Execution Order

1. Inspect the current repository, schema and existing branch/inventory/accounting architecture.
2. Run the existing test suite and establish a baseline.
3. Map current data and design the location hierarchy.
4. Design safe migrations and test them against realistic existing data.
5. Implement branch and warehouse management.
6. Implement location-aware inventory.
7. Implement transfer workflow and idempotency.
8. Implement central purchasing/replenishment integration.
9. Implement branch financial and business reporting.
10. Implement lightweight staff controls and approval workflows.
11. Extend the Advanced Business Command Center.
12. Add comprehensive automated tests.
13. Run the full regression suite.
14. Perform security, tenant-isolation, data-integrity and performance review.
15. Document the final implementation and known limitations.

Final Instruction to Claude

Build Phase 6 as a safe, production-quality incremental upgrade. Do not rebuild the existing core and do not reset the database. The architecture must support both a small single-branch shop and a larger multi-branch organization without unnecessary complexity. Every location-scoped operation must enforce tenant and branch authorization at the backend. Stock transfers must be transaction-safe, auditable and idempotent. Integrate location data into accounting, inventory, procurement, reporting and the Advanced Owner Command Center. Do not add a heavy HR/payroll module; keep staff functionality focused on business control, permissions, performance and approvals. Preserve existing data, offline/sync behavior and backward compatibility. The result must provide a stable foundation for Phase 7: Advanced Optical + Eye Clinic + CRM.